SHAMUS MURPHY

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EDUCATION

- **University of Vermont** — Bachelor of Science in Mathematics, Minor in Computer Science
- *Expected Graduation: May 2027, GPA 3.25*
- **Relevant Coursework:** Data Structures and Algorithms, Data Privacy, Cybersecurity Principles, Computability and Complexity, Abstract Algebra, Number Theory

PROJECTS

Network Intrusion Detection with Random Forest and Gradient Boosting on the CIC-IDS2017 Dataset *(CS 3870 – Data Science Pinnacle | Python, scikit-learn, LightGBM, Pandas)*

• Led a team of three building an intrusion detection system on 2.2 million network flows, removing 82K duplicate records, handling infinite flow-rate values and reducing redundant features (78 raw columns to 60 features)

• Trained a random forest for binary attack detection (0.97 attack recall, 1.00 precision on 446k held-out flows) and analyzed the traffic features driving the splits

• Built a weighted LightGBM classifier for 14-way attack identification with 0.97-1.00 F1 on Dos, DDos, and FTP/SSH brute-force attacks

Privacy and Utility Tradeoffs in Medical Risk Prediction *(CS 3110 – Data Privacy) | Python, PyTorch, Opacus)*

- Trained a differentially private neural network to predict diabetes on a 100k record medical dataset using DP-SGD (Opacus) at δ=1e-5, after removing leakage-prone target columns
- Swept the noise multiplier from 0.45 - 1.5 (ε from 9.3 - 0.33) and benchmarked against a non-private baseline of 90.4% accuracy.
- Retained 98% of baseline accuracy at ε = 1.45 (88.6%) and identified a sharp drop in utility below ε = 0.5 (79.5% at ε = 0.33)

RESEARCH EXPERIENCE

Research Experience for Undergraduates (REU) — Cryptographic Protocols *January 2026 – May 2026*

- Implemented a secure aggregation protocol in PyChor, a choreographic programming library, letting multiple parties compute a sum without revealing individual inputs
- Verified protocol correctness up to the client-dropout phase using choreography-based abstractions for secure distributed computation

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, SQL (MySQL)

Machine Learning: PyTorch, scikit-learn, LightGBM, Opacus

Data: NumPy, Pandas, Matplotlib, Seaborn

Security: Differential Privacy, Secure Aggregation, Cryptography

Development: Git, Linux, CLI

WORK EXPERIENCE

Line Cook — Taco Gordo, Burlington, VT *March 2026 – Present*

- Prepare and plate high-volume food orders across grill, sauté, and prep stations in a fast-paced kitchen environment

Bar Back — Rooney's Oceanfront Restaurant, Long Branch, NJ *May 2024 – August 2025*

- Supported high-volume service operations in a fast-paced, 300+ seat oceanfront restaurant, coordinating with bartenders and wait staff to maintain efficient drink and supply flow during peak service